Tarek Alam Bhuiyan

Machine Learning | Data Science | Software Engineering

tarekalambhuiyan@gmail.com | Mohammadpur, Dhaka - 1207 | +8801601070407 | Linkedin | Github | LeetCode

Profile

Computer Science undergraduate with hands-on experience in Machine Learning, Data Science, Data Analytics & Annotation, Natural Language Processing, and full-stack development. Skilled in Python, JavaScript, and SQL, with proven ability to develop ML pipelines, implement deep learning architectures, and create scalable web solutions. Committed to continuous learning and applying data-driven approaches to deliver impactful solutions to real-world problems.

Professional Experience

Trainee Technology Specialist - Software, Data & AI Solutions 22/02/2026 – Current
Betopia Limited Dhaka, Bangladesh
Selected among the top 40 candidate for the Betopia Tech Recruitment Program (BTRP)

Education

BSc. in Computer Science And Enginering 2022 – 2026
BRAC University Dhaka, Bangladesh

Projects

MycoPredict - Numpy, Pandas, Matplotlib, Seaborn, Scikit-learn
Mushroom Edibility Classification using Machine Learning

- Leveraged the UCI dataset (8,124 samples, 22 features) to predict mushroom edibilty.
- Preprocessed data, handled missing values, removed redundant features, applied label encoding.
- Conducted EDA and feature analysis to identify key predictors (gill-size, population, habitat).
- Trained multiple ML models and evaluated performance using accuracy, F1-score, recall, AUC.

Word2Sense - NumPy, Pandas, Matplotlib, TensorFlow/Keras, Scikit-learn
Comparative Study of Word Representations with ML and Neural Models

- Built a large-scale text classification system on ~340,000 question–answer pairs across 10 categories
- Compared embeddings (BoW, TF–IDF, GloVe, Skip-gram) across multiple ML and neural models.
- Best results: Logistic Regression + TF–IDF (Acc: 0.703) and BiLSTM + Skip-gram (Acc: 0.725).
- Used hyperparameter tuning, dropout, early stopping; evaluated with accuracy and F1-score.

PetSphere - Node.js , Express.js, Tailwind CSS, MongoDB
A Comprehensive Pet Adoption and Management System

- Built required features for pet adoption, health tracking and monitoring.
- Implemented REST APIs, JWT authentication, and role-based dashboards.
- Implemented real-time chat between users using Socket.IO .
- Collaborated in agile 4-member team using Git/GitHub.

Skills

Programming Languages
Python, C, C++, Javascript, SQL

Tools and Technologies
CVAT, Excel, MySQL, MongoDB, Git, Github, Figma

Libraries & Frameworks
Numpy, Pandas, Matplotlib, Seaborn, Scikit-learn, NLTK, Gensim, TensorFlow, Keras, Node.js, Express.js, Tailwind CSS

Certificates

AI Fundamentals — Datacamp

Intermediate SQL — Datacamp